Creation of Redshift Tables and Execution of Airflow Workflow

Summary of the Lab

In this lab, required databases, schemas, and tables are created in Amazon Redshift to store the final processed output. An Apache Airflow DAG is then deployed and executed to validate, transform, and load data into Redshift. The lab also includes configuring IAM permissions, uploading input datasets to Amazon S3, creating a Redshift connection in Airflow, and validating the final output in Redshift tables. Successful execution of the workflow confirms end-to-end data processing using Airflow and Redshift.

Prerequisite for the Lab

o Amazon Redshift Serverless environment created

o AWS Managed Apache Airflow environment running

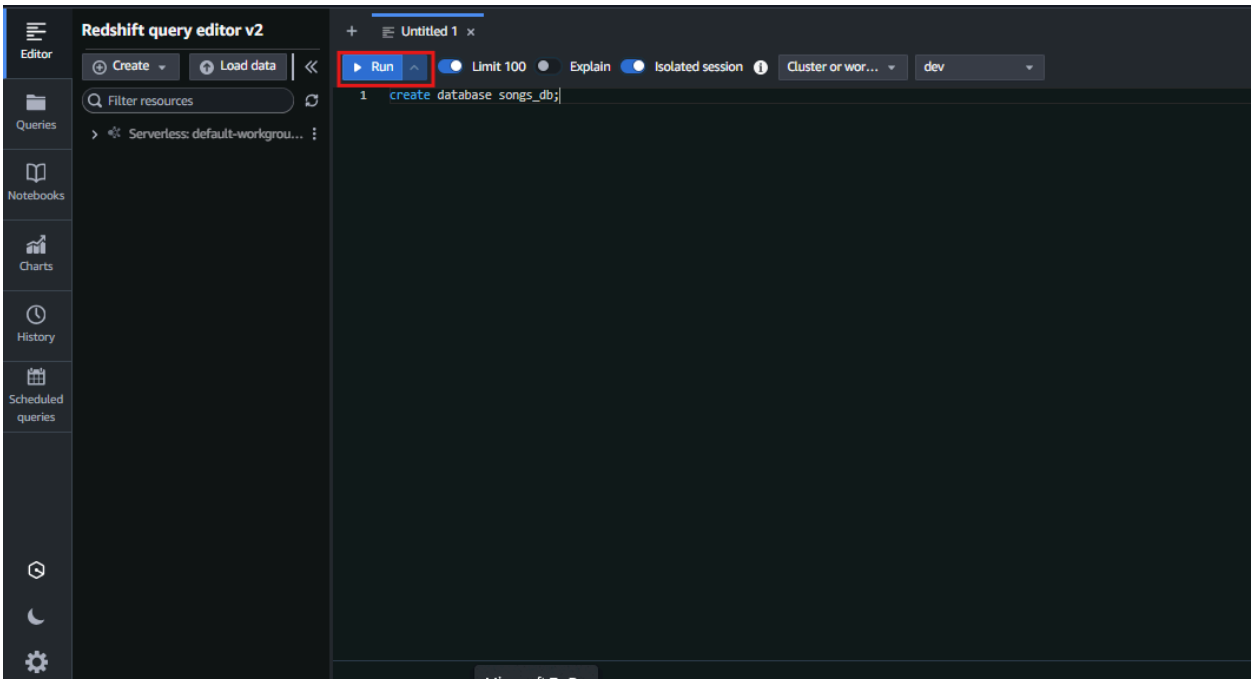
o Access to Redshift Query Editor v2

o Airflow DAG script available

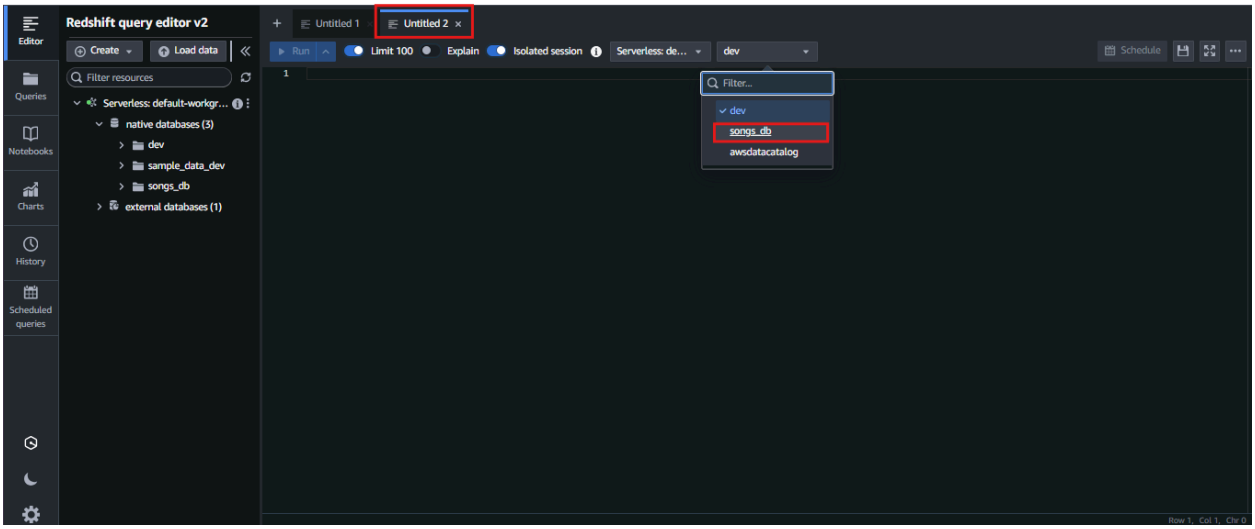
o Input CSV files (songs.csv, users.csv, streams files)

Procedure

- Open the SQL file named `redshift_tables.sql`
- Execute the SQL statement to create a database named `songs_db`



- Open a new SQL editor and select the `songs_db` database



- Execute the remaining SQL statements to perform the following:

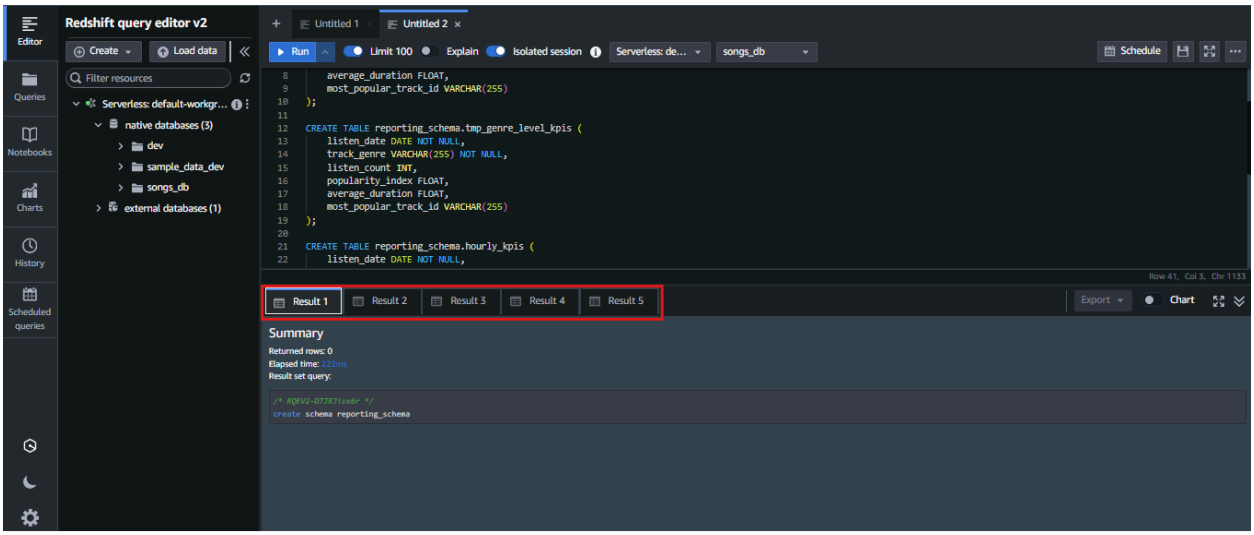
o Create a schema named `reporting_schema`

o Create four tables inside the schema

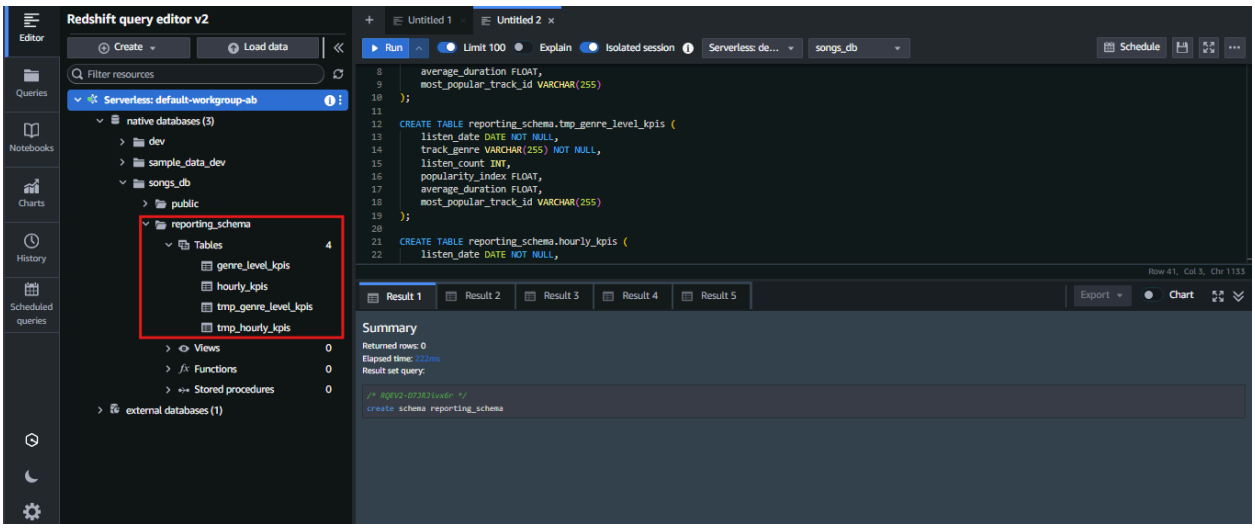
▪ genre_level_kpis

▪ temp_genre_level_kpis

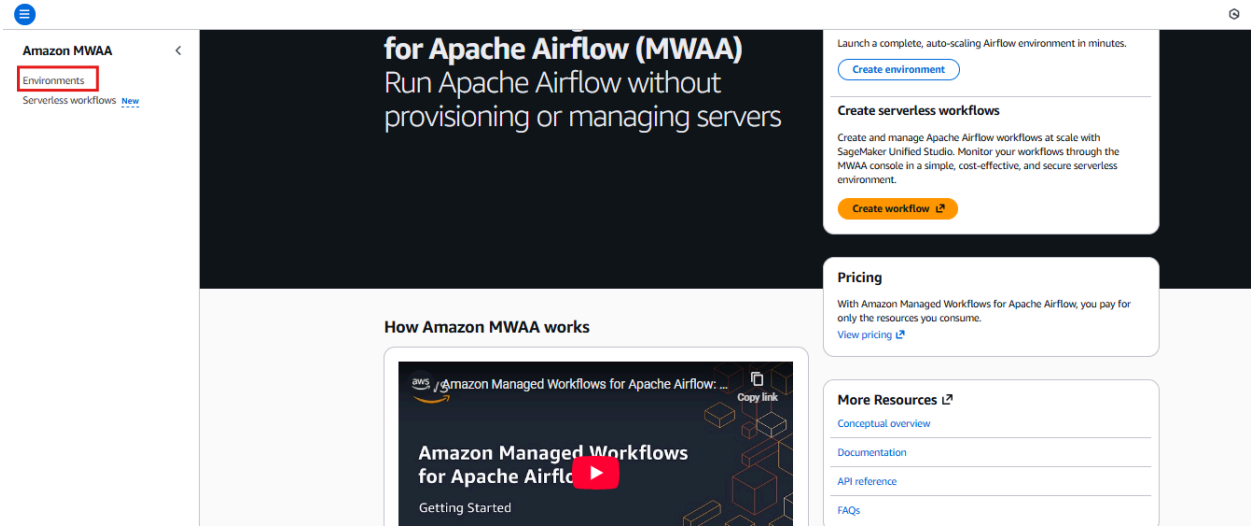
- only_kpis
- temp_only_kpis



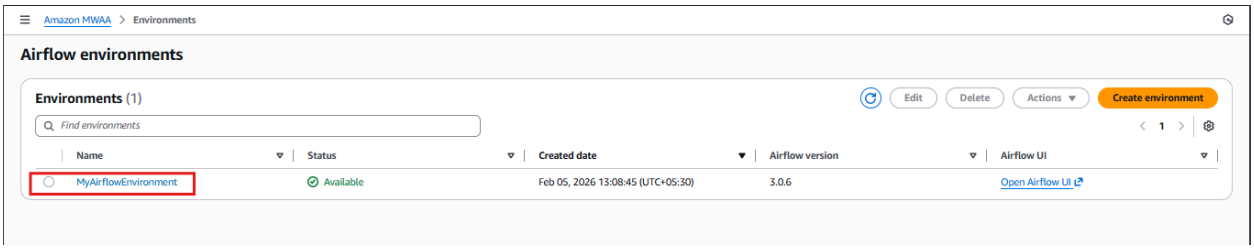
- Refresh the database to verify that the schema and tables are created successfully



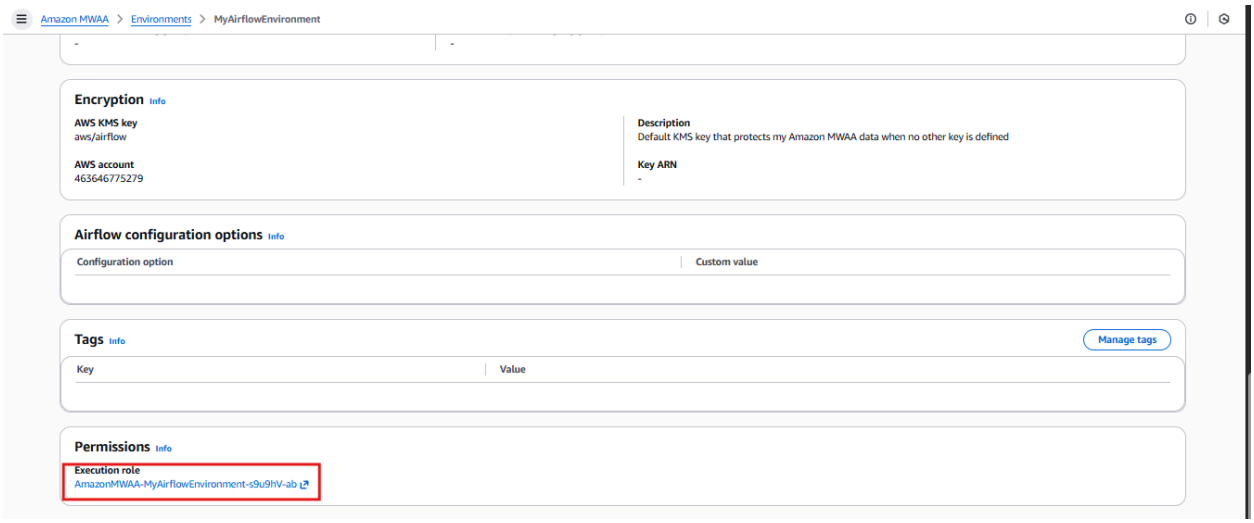
- Navigate to the AWS Console → Managed Workflows for Apache Airflow



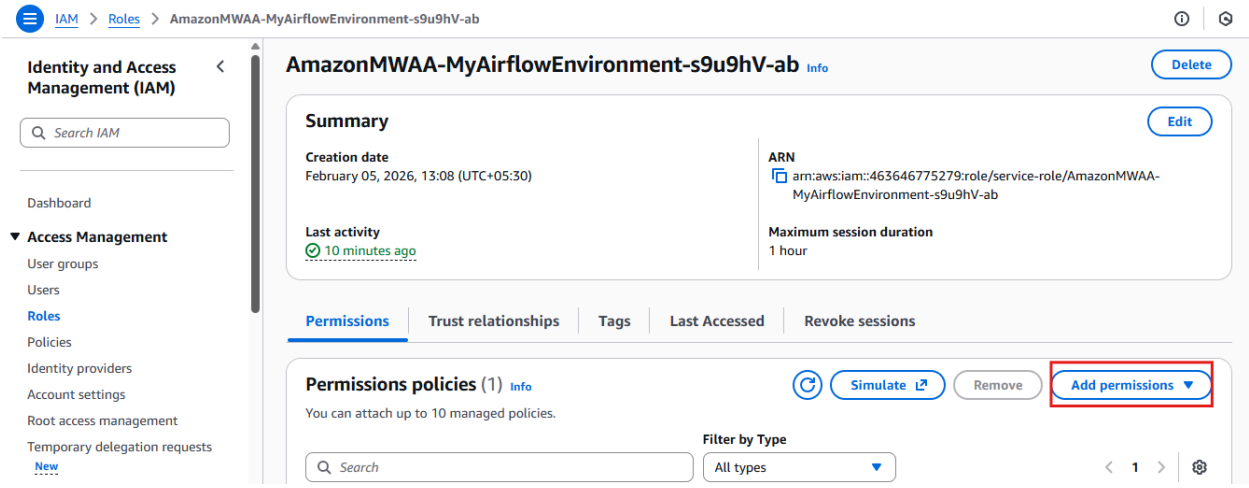
- Open the Airflow environment



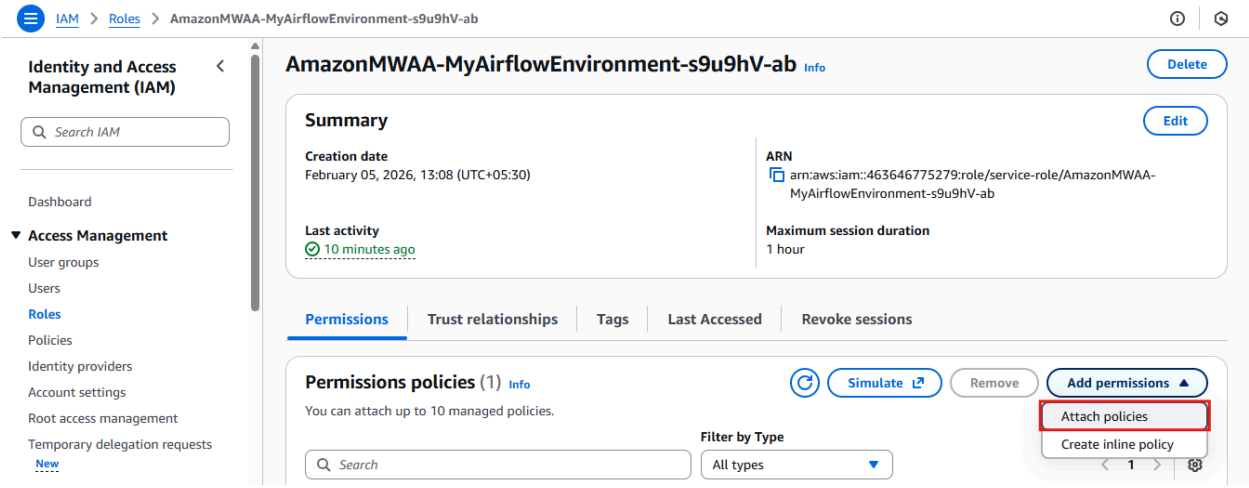
- Scroll to the Permissions section and click on the Execution IAM Role

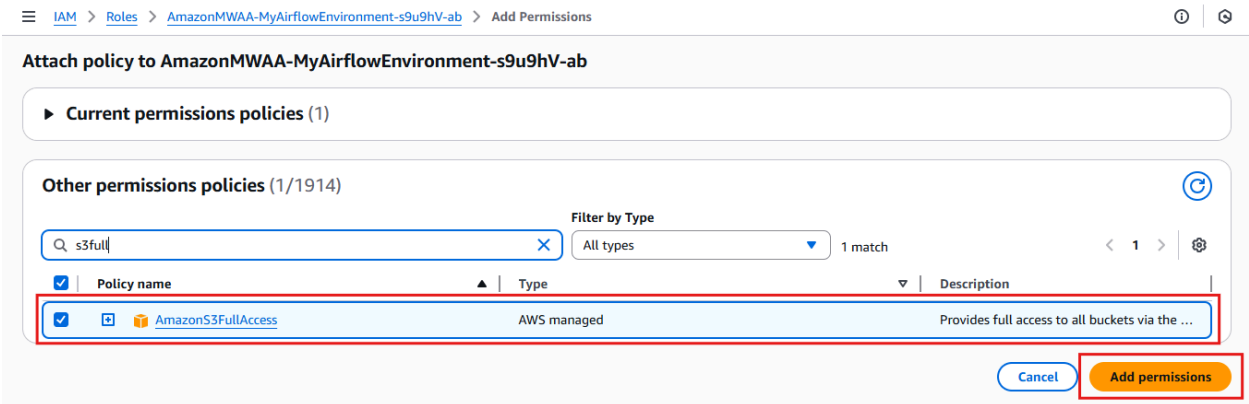


- Add a new permission policy to the role

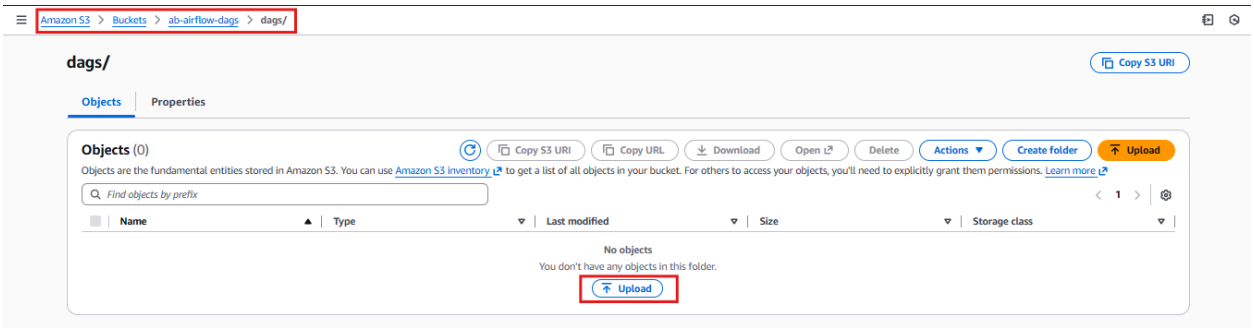


o Attach the policy **AmazonS3FullAccess**

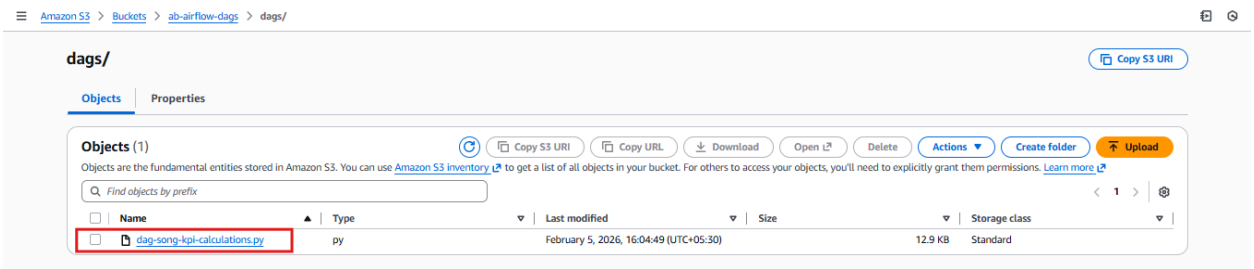




- Go to the S3 bucket created for Airflow DAGs

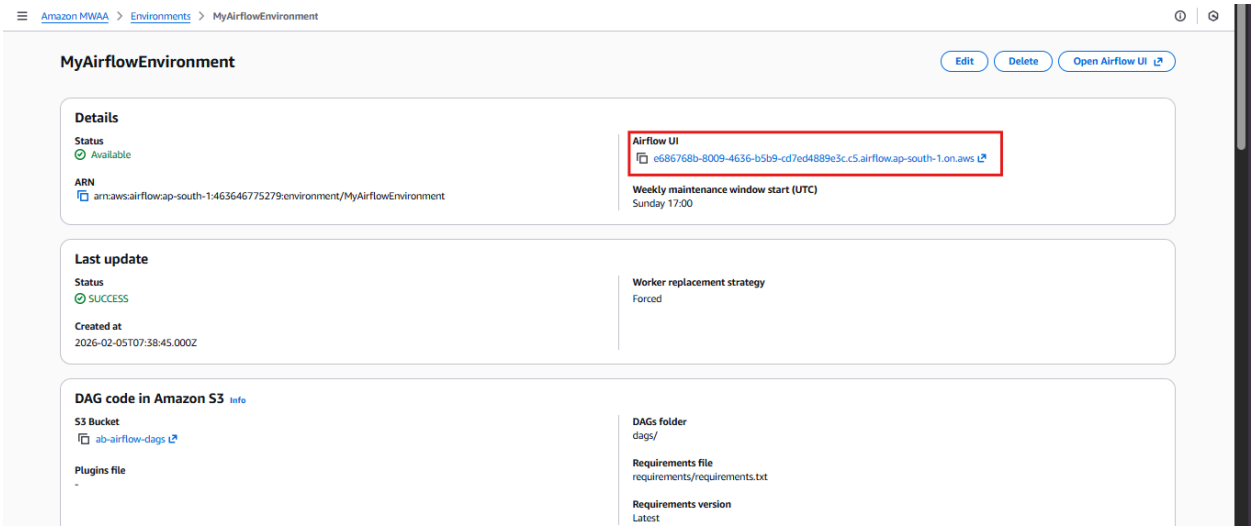


- Upload the Airflow DAG script into the `dags` folder

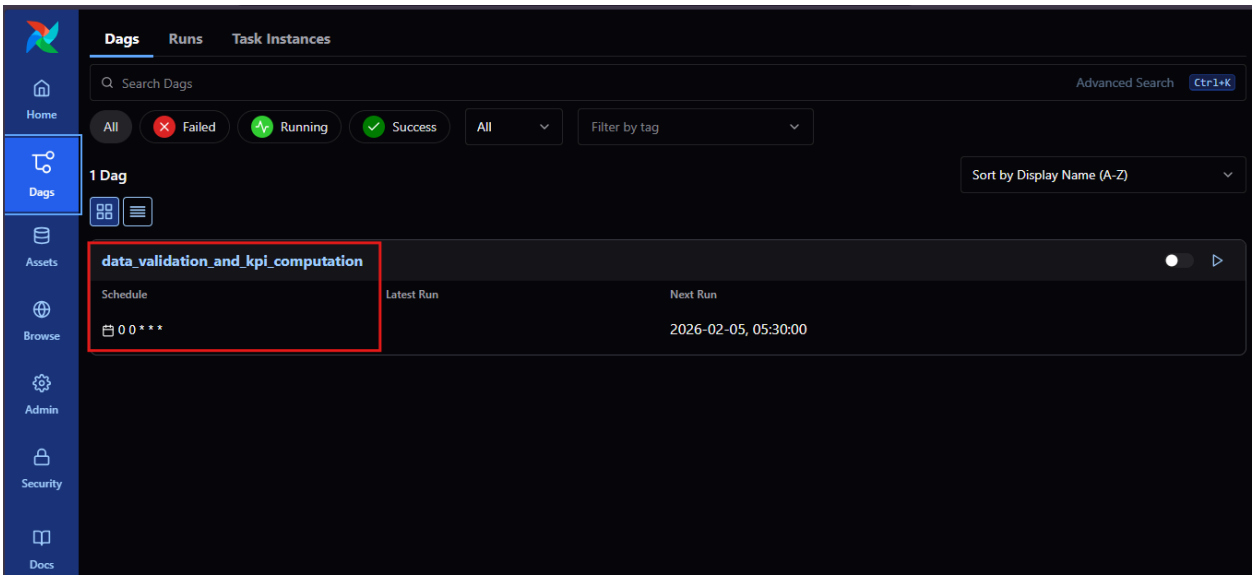


o Ensure the S3 bucket name inside the DAG script matches your bucket name

- Open the Airflow UI from the MWAA console



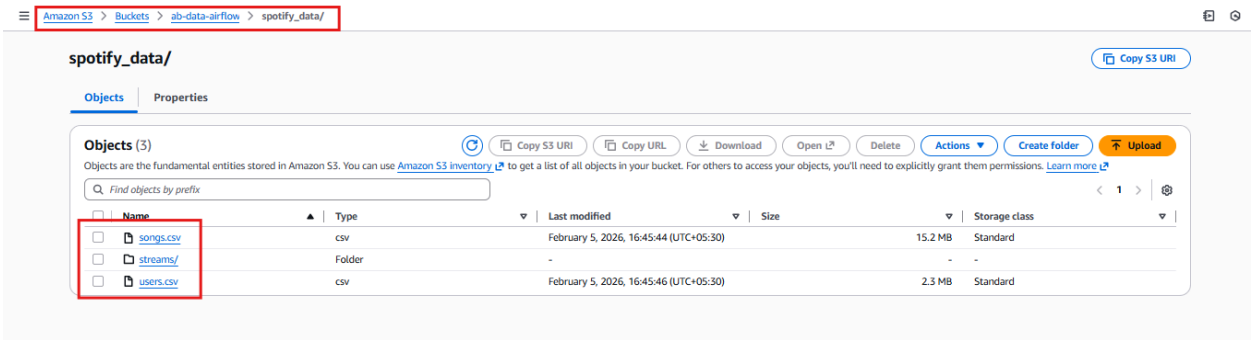
- Wait a few minutes for the DAG to appear in the UI



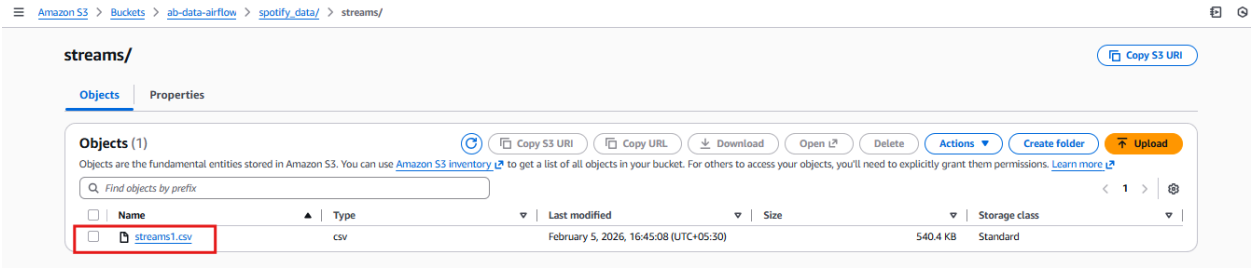
- Create a new S3 bucket (different from the Airflow bucket) for input data
- Inside the bucket, create a folder named `spotify_data`
- Upload the following files inside `spotify_data`:

o songs.csv

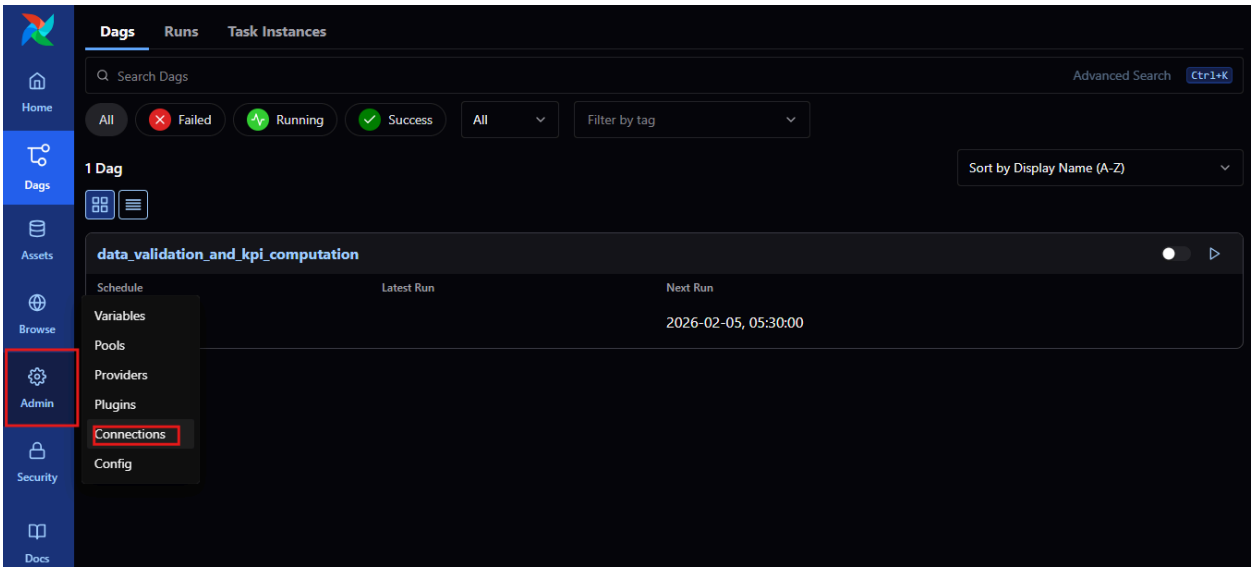
o users.csv



- Create a subfolder named `streams` inside `spotify_data`
- Upload any one of the stream files (streams_1, streams_2, or streams_3) into the streams folder



- In the Airflow UI, go to Admin → Connections



- Click on the plus (+) icon to create a new connection

- Set the connection details as follows:

o Connection ID: redshift_default

o Connection Type: Redshift

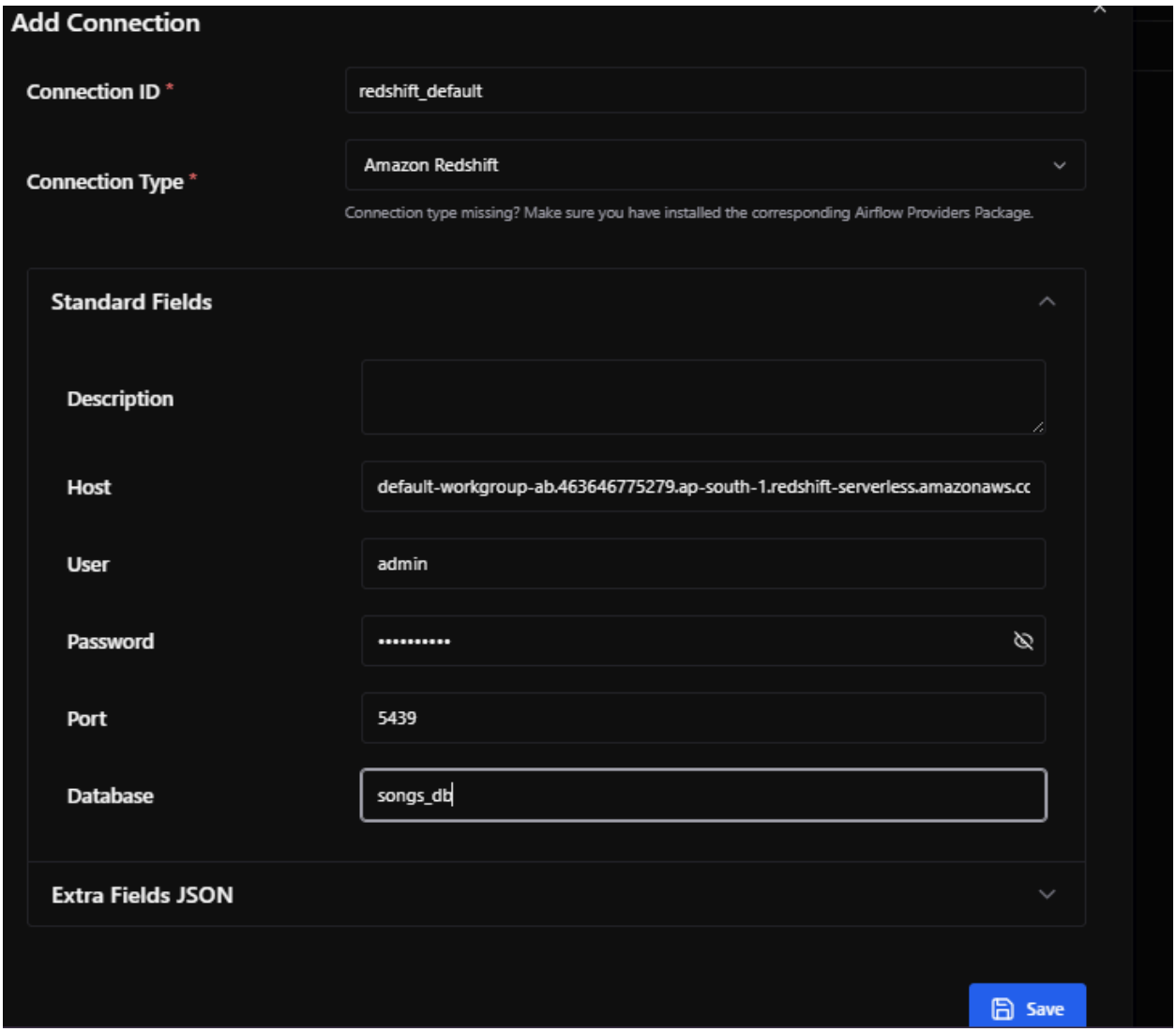
o Host: Redshift Serverless endpoint (without database name and port)

o Schema: songs_db

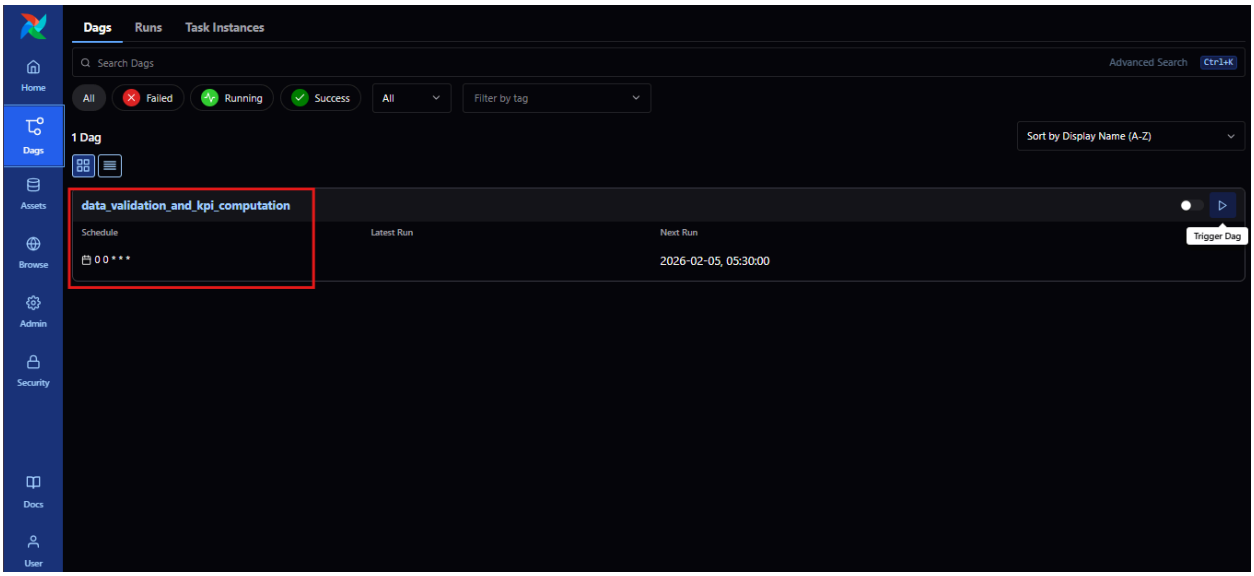
o Login: admin

o Password: Redshift admin password

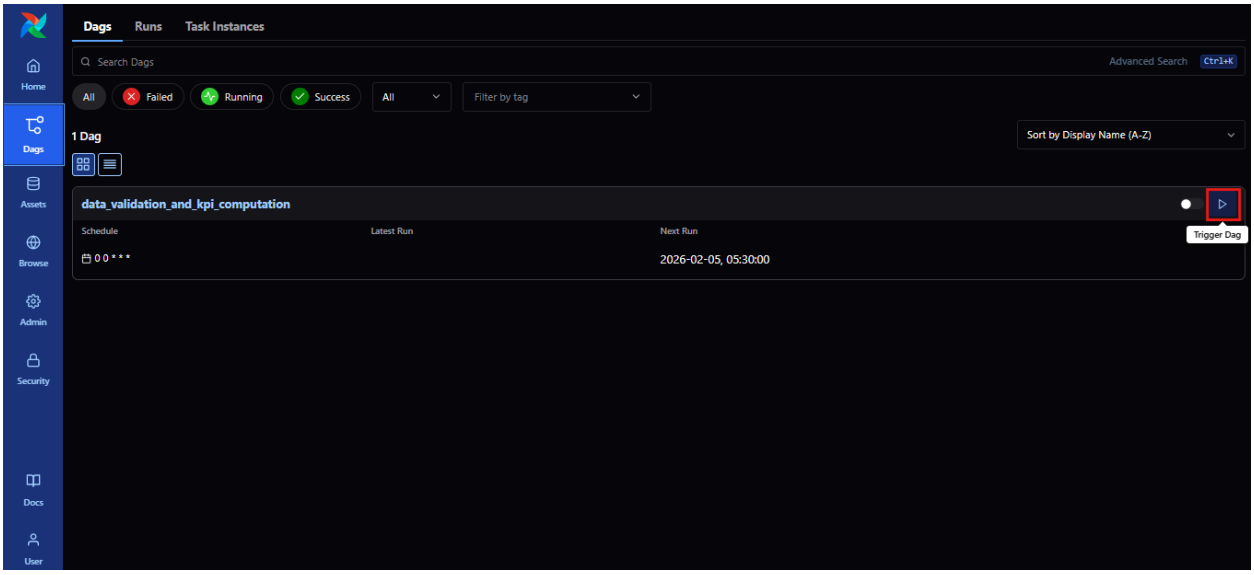
o Port: 5439



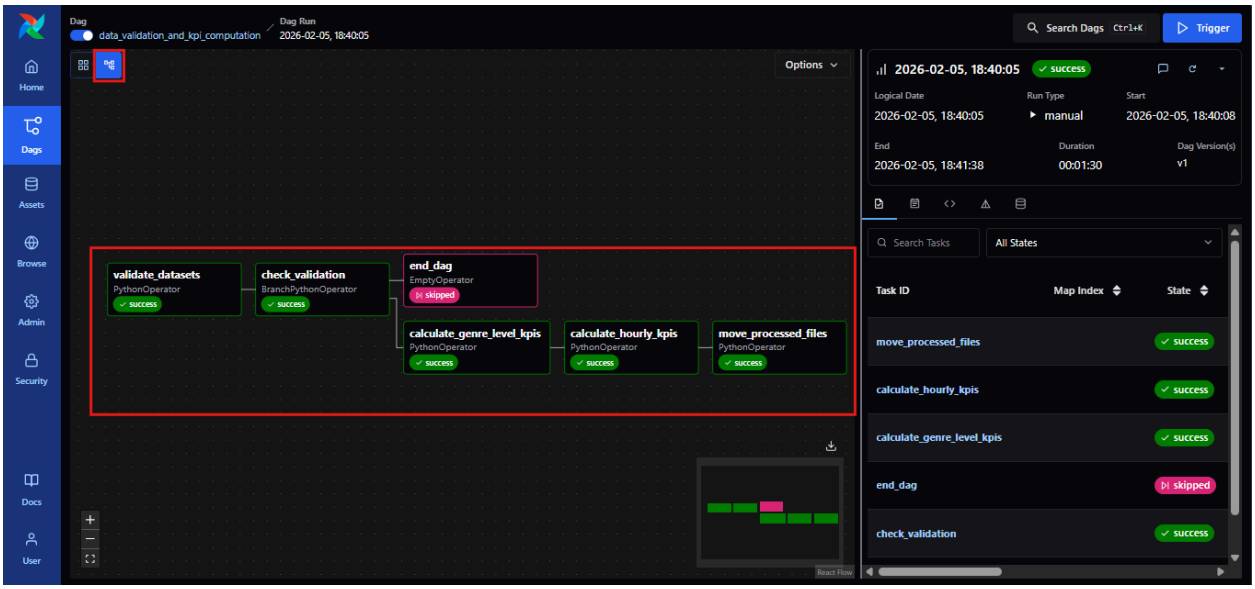
- Save the connection
- Go to the DAGs section in Airflow UI



- Click on the play icon to trigger the DAG execution

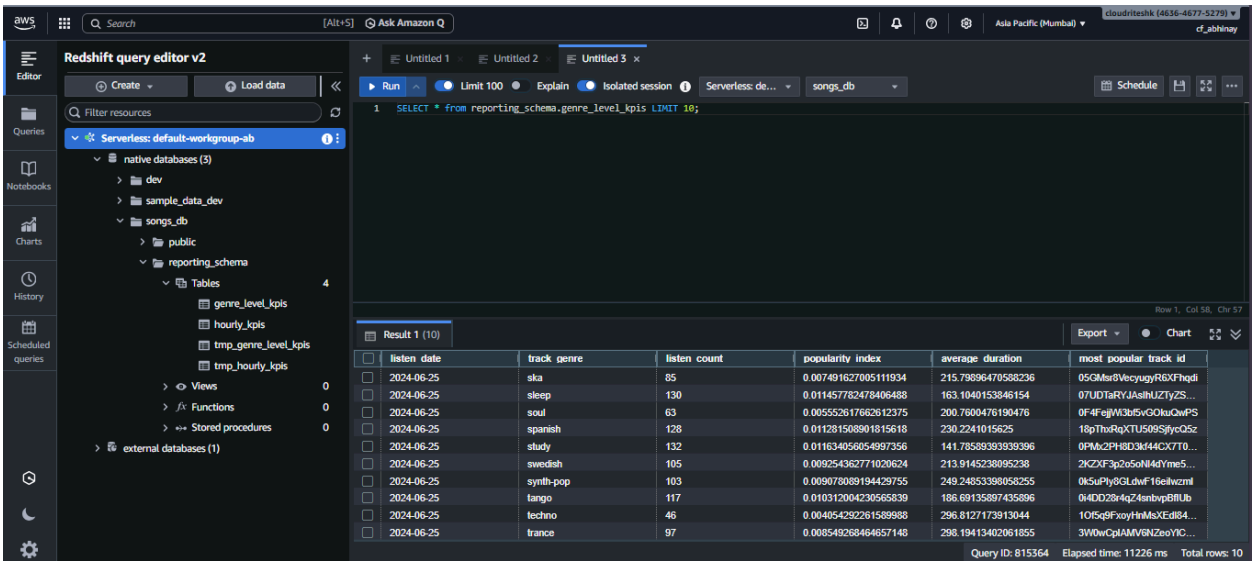


- Open the DAG and view the Graph view to observe task execution flow
- Wait for 2–3 minutes for the workflow to complete
- Verify that all tasks are marked successful (green)

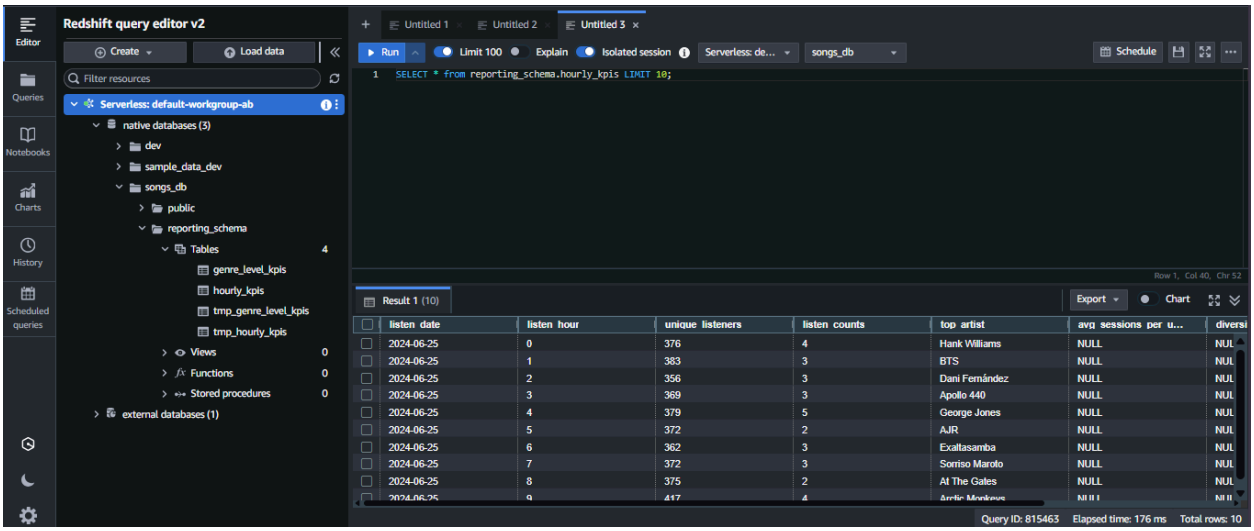


- Review task logs for validation and debugging if required
- Go back to Redshift Query Editor
- Run the following queries to validate data insertion:

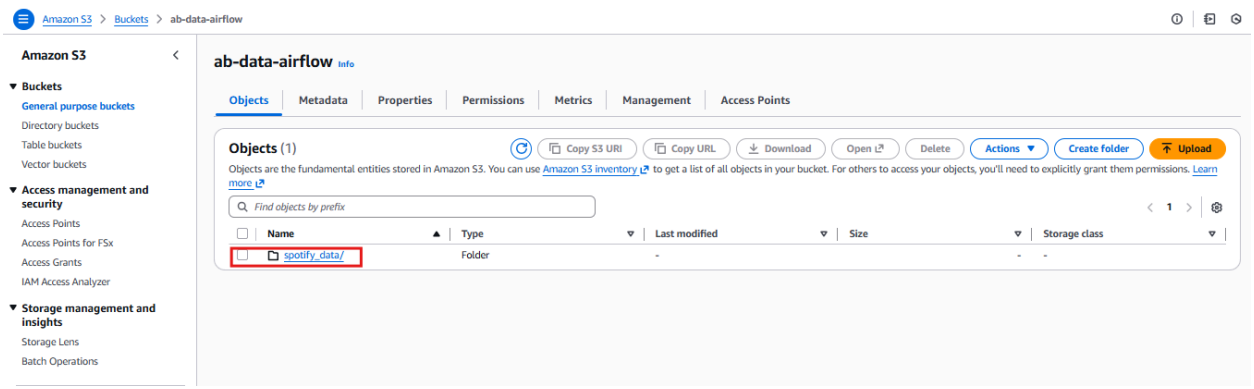
o `SELECT * FROM genre_level_kpis LIMIT 10;`



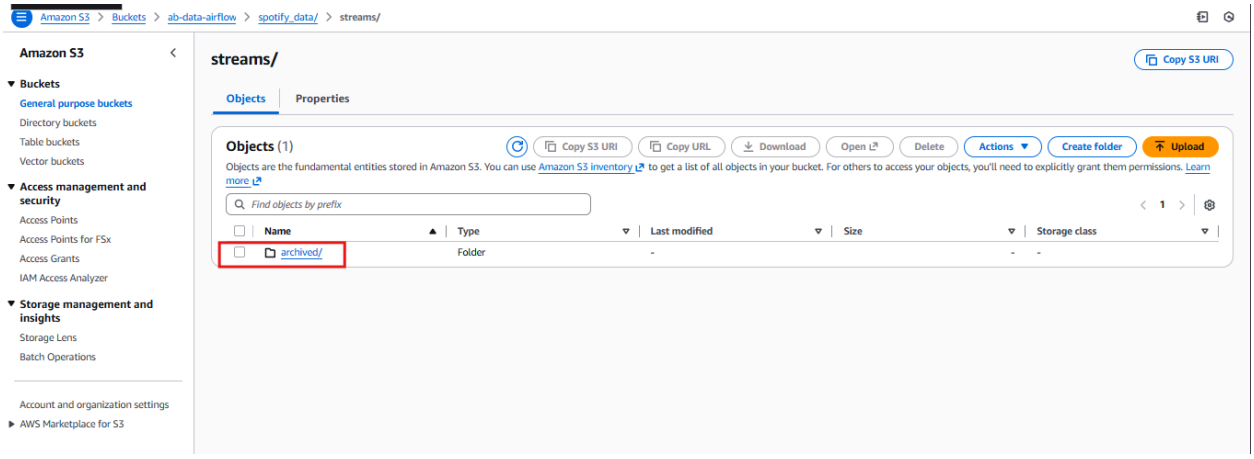
o `SELECT * FROM only_kpis LIMIT 10;`



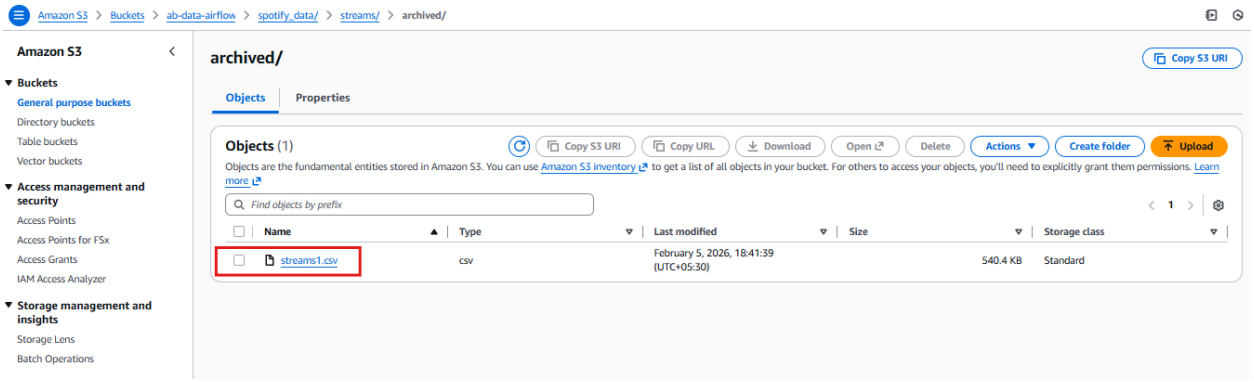
- Verify that the transformed data is successfully loaded into Redshift tables
- Navigate to the S3 bucket containing `spotify_data`



- Open the `streams` folder and verify that an `archived` folder is created



- Confirm that the processed stream file is moved to the archived folder



Result

The Redshift database, schema, and tables were successfully created. The Airflow DAG executed without errors, validated and transformed the input data, and loaded the final output into Redshift tables. Processed stream files were archived in Amazon S3. This confirms successful execution of the first end-to-end workflow using Apache Airflow and Amazon Redshift.